

The Blind Spot Paradox

The Blind Spot Paradox

When a self-repairing model destroys the evidence its own monitor needs

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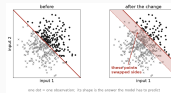
The Blind Spot Paradox: when a self-repairing model destroys the evidence its own monitor needs.

The Blind Spot Paradox

└ Concept drift: the rule changes, the data does not

First, the phenomenon. A model learns a rule: this input goes with that answer. Then the world changes and the rule no longer holds. That is concept drift, and here is the cleanest version of it, the one we simulate. Two inputs, a straight boundary, a label on each side. The change moves the boundary, and nothing else. Only the shaded band has swapped sides, and a model trained before is now wrong on it.

Concept drift: the rule changes, the data does not



A model trained on the left keeps predicting, confidently, and is now **wrong** on the shaded band.

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└ Defence one: a model that repairs itself

The first defence repairs itself. Its building block is a decision tree: it splits the picture with simple cuts, and answers by asking which piece a new point falls into. One tree is fragile, so ten of them vote. That is the Adaptive Random Forest, the model we study. A tree whose answers start going wrong is thrown away, and a fresh one grows in its place. Ten trees, replaced one at a time.

- A **decision tree** splits the picture with simple cuts, and answers by asking which piece a new point falls into.
- One tree is fragile, so ten of them **vote**. That is the Adaptive Random Forest, the model we study.
- **It repairs itself**: a tree whose answers start going wrong is thrown away, and a fresh one grows in its place.

Ten trees, replaced one at a time. Remember that number.

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└ Defence two: a watchdog that raises the alarm

The second defence comes from a different community. A watchdog reads the model's mistakes, one observation at a time, and adds up the excess. Once that pile crosses a threshold it calls a human: audit, retraining, pause. That alarm is the only thing a human ever sees. Two communities, two tools, never measured against each other.

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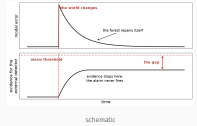
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└ Put them together, and they cancel out

Put the two together, and they cancel out. The boundary moves, the mistakes begin: top curve. The watchdog needs exactly those mistakes. But the forest repairs itself, the mistakes stop, and the pile stops growing before it is high enough. The repair erased the evidence. The better the model fixes itself, the more surely the alarm stays silent.

Put them together, and they cancel out

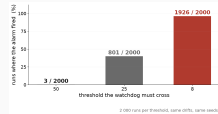


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└ Why not just lower the threshold?

Does the alarm ever fire? That depends on the threshold. We re-ran the benchmark at three settings, two thousand runs each. At the highest it fired three times out of two thousand. Lower it, and it fires almost every time. So why not lower it? Because a low threshold also fires when nothing happened, and an alarm that cries wolf gets switched off. The setting quiet enough to be trusted is the one that never sees the drift.

Why not just lower the threshold?



A low threshold also fires when **nothing has happened**, and an alarm that cries wolf gets switched off.

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└ Our contribution: the accepted clock is wrong

To study a race you need a stopwatch, and that is where we found something. The date the literature uses to say the model has recovered is the moment the very first tree is replaced: one out of ten. Wait until three quarters of the forest has been renewed, and you wait roughly ten times longer. That stopwatch does not time a repair. It times the start of one.

Our contribution: the accepted clock is wrong

called "repaired"
by the literature

86

three quarters of
the forest renewed

1225

0 200 400 600 800 1000 1200 1400
observations seen since the change

median of 100 runs, forest of 10 trees, change size 0.24

It marks the first tree out of ten, not a repaired forest.

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└─Where we stand

Where we stand. The paradox is real and reproducible. The clock the field relies on is biased, always the same way, so every conclusion drawn from it inherits that bias. What remains is to turn this into advice a practitioner can use. Thank you.

- The paradox is **real and reproducible**, and we can say when the evidence falls short of what the alarm needs.
- The clock the field relies on is **biased, always the same way**. Every conclusion drawn from it inherits that bias.
- **Next:** watch the model and the watchdog on one common clock, and turn this into advice a practitioner can use.

Thank you.